

Part B – Weekly Capstone Assignment Template

This template is used for **each of the 14 Part B capstone submissions** throughout the semester. The structure remains the same every week so that you can focus on *thinking and judgment*, not guessing expectations.

Your goal is not to achieve the best performance, but to **reason carefully about how this week's machine learning technique applies (or does not apply) to your project**.

1. Project Context (Brief)

Provide the same short context each week so graders can orient quickly.

- **Project Title: MLB Home Runs**
 - ****Data Modality: Tabular **** (tabular, image, audio, text, other)
 - **Task Type: Regression** (regression, classification, clustering, sequence modeling, etc.)
 - **One-Sentence Goal: Predict how many home runs MLB players will hit next season.** What are you trying to predict, detect, or understand?
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2. This Week's Technique and Its Assumptions

- **Technique / Model Family Covered This Week: Practical Deep Learning and Frameworks**
- ****Key Assumptions of This Technique:**
 - These techniques are useful to my case.** (1–2 bullets)

Fit Assessment (required): Practical Deep Learning Systems

I expect this technique to be a **partial** fit for my project because: Several points would be useful or probably useful, like checkpointing. Others, like dataloader would not hurt, but my model is so small it might not actually make much difference. Overall, I will try applying what seems potentially useful. (2–4 sentences)

3. Representation or Proxy Used

Describe how your data was represented so that this week's technique could be applied.

Examples include:

- Hand-engineered features
- Summary statistics
- Frozen embeddings
- Dimensionality reduction

- A proxy task
 - **Representation or Proxy Chosen: Maintain at Hand-engineered features applied to MLP.**
 - **Why this representation was reasonable for this week: I'm simply adding a few extra bits to the MLP set from 11, so it should stay the same. w**
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4. What Was Attempted

Be concrete and scoped. Do not list everything you *could* have done.

- What you implemented this week: Added random seeds before running MLP. Added checkpointing after you run the MLP.
- What you intentionally did *not* attempt and why: Skipped dataloading and of course CNN, as this is tabular data and only a few thousand rows.
- Any constraints encountered (data, labels, compute, time): None.

5. Results or Observations

You may include metrics **if applicable**, but qualitative observations are also valid.

Examples:

- Evaluation metrics
- Training behavior or convergence issues
- Error patterns
- Unexpected behaviors

Checkpoint worked as expected/hoped and confirmed to be successful.

6. Interpretation and Judgment

This section matters most.

Reflect on:

- Why the method behaved as it did
- Which assumptions held or failed
- What this reveals about your data or problem framing

(1–2 thoughtful paragraphs)

Some of these additions that could have been implemented show that this is a unique case, in that it's so small and honestly a weird type of data, so it is a bit different from many models would be. Also of course, something like a visual analysis would be much different and probbaly far more complex than a tabular data analysis, utilizing things like CNN. Additionally, the checkpoint shows the MLP should be good, since it gave the same output.

7. Forward-Looking Adjustment

Answer **one** of the following:

- What will you keep, change, or discard before the next assignment?
 - I did not get to adding a secondary threshold yet.
 - What would you try next if data or resources were not constrained?
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8. Mismatch Acknowledgment (Complete Only If Applicable) -

If this week's technique was a poor fit, explain:

- Why it does not align with your project: not a poor fit, only partial.
 - Evidence supporting that conclusion: CNN was not applicable, but the others were at least potentially useful.
 - What value this attempt still provided: Showed variance between different projects and models, especially the difference that having a small model can cause.
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Submission Notes

- Written submission format: **Markdown or PDF**
- Code or notebooks: **optional unless explicitly requested**
- Performance is **not** graded competitively
- Clear reasoning and honest reflection matter more than results